

**Instructions.** (15 points) Sections 15.7, 15.8, 16.1, 16.2. Show relevant work for full credit.

- (4<sup>pts</sup>) **1.** Evaluate the integral  $\iiint_E z \sqrt{x^2 + y^2} dV$ , where  $E$  is the solid inside the cylinder  $x^2 + y^2 = 1$ , between  $z = 0$  and the cone  $z = \sqrt{x^2 + y^2}$ .

**Solution:**

Convert to cylindrical:  $\sqrt{x^2 + y^2} = r$ ,  $dV = r dz dr d\theta$ .

The region:  $0 \leq \theta \leq 2\pi$ ,  $0 \leq r \leq 1$ ,  $0 \leq z \leq r$ .

The integrand becomes  $z \cdot r$ , so:

$$\int_0^{2\pi} \int_0^1 \int_0^r z \cdot r \cdot r dz dr d\theta = \int_0^{2\pi} \int_0^1 \int_0^r z r^2 dz dr d\theta$$

**Inner integral ( $dz$ ):**

$$\int_0^r z r^2 dz = r^2 \cdot \frac{z^2}{2} \Big|_0^r = \frac{r^4}{2}$$

**Middle integral ( $dr$ ):**

$$\int_0^1 \frac{r^4}{2} dr = \frac{1}{2} \cdot \frac{r^5}{5} \Big|_0^1 = \frac{1}{10}$$

**Outer integral ( $d\theta$ ):**

$$\int_0^{2\pi} \frac{1}{10} d\theta = \boxed{\frac{\pi}{5}}$$

- (4<sup>pts</sup>) **2.** Evaluate the integral  $\iiint_E z dV$ , where  $E$  is the region inside the sphere  $x^2 + y^2 + z^2 = 4$  and above the cone  $\phi = \pi/3$ .

**Solution:**

Convert to spherical:  $z = \rho \cos \phi$ ,  $dV = \rho^2 \sin \phi d\rho d\phi d\theta$ .

The region:  $0 \leq \theta \leq 2\pi$ ,  $0 \leq \phi \leq \pi/3$ ,  $0 \leq \rho \leq 2$ .

$$\int_0^{2\pi} \int_0^{\pi/3} \int_0^2 \rho \cos \phi \cdot \rho^2 \sin \phi d\rho d\phi d\theta = \int_0^{2\pi} \int_0^{\pi/3} \int_0^2 \rho^3 \cos \phi \sin \phi d\rho d\phi d\theta$$

**Inner integral ( $d\rho$ ):**

$$\int_0^2 \rho^3 d\rho = \frac{\rho^4}{4} \Big|_0^2 = 4$$

**Middle integral ( $d\phi$ ):** Let  $u = \sin \phi$ ,  $du = \cos \phi d\phi$ :

$$\int_0^{\pi/3} \cos \phi \sin \phi d\phi = \frac{\sin^2 \phi}{2} \Big|_0^{\pi/3} = \frac{(\sqrt{3}/2)^2}{2} = \frac{3}{8}$$

**Outer integral ( $d\theta$ ):**

$$\int_0^{2\pi} 4 \cdot \frac{3}{8} d\theta = \frac{3}{2} \cdot 2\pi = \boxed{3\pi}$$

- (3<sup>pts</sup>) **3.** (a) (2 pts) Compute  $\nabla f$  for  $f(x, y) = xe^y + y^2$ .

**Solution:**

$$\nabla f = \langle f_x, f_y \rangle = \langle e^y, xe^y + 2y \rangle$$

- (b) (1 pts) Using part (a), explain why the vector field  $\mathbf{F}(x, y) = \langle e^y, xe^y + 2y \rangle$  is conservative.

**Solution:**

Since  $\nabla f = \langle e^y, xe^y + 2y \rangle = \mathbf{F}$ , the field  $\mathbf{F}$  is a gradient field. By definition, a vector field that equals  $\nabla f$  for some function  $f$  is **conservative**.

- (4<sup>pts</sup>) **4.** Evaluate the line integral  $\int_C \mathbf{F} \cdot d\mathbf{r}$ , where  $\mathbf{F}(x, y) = \langle x + y, y \rangle$  and  $C$  is the line segment from  $(1, 0)$  to  $(3, 2)$ .

**Solution:**

**Parameterize:**  $\mathbf{r}(t) = \langle 1, 0 \rangle + t\langle 2, 2 \rangle = \langle 1 + 2t, 2t \rangle$ ,  $0 \leq t \leq 1$ .

$\mathbf{r}'(t) = \langle 2, 2 \rangle$ .

**Compute  $\mathbf{F}(\mathbf{r}(t))$ :**

$$\mathbf{F} = \langle (1 + 2t) + 2t, 2t \rangle = \langle 1 + 4t, 2t \rangle$$

**Dot product:**

$$\mathbf{F} \cdot \mathbf{r}'(t) = (1 + 4t)(2) + (2t)(2) = 2 + 8t + 4t = 2 + 12t$$

**Integrate:**

$$\int_0^1 (2 + 12t) dt = [2t + 6t^2]_0^1 = 2 + 6 = \boxed{8}$$